Strategic Asset Allocation

Northbridge University Endowment · Recommendation to the Investment Committee

Recommendation: adopt the Growth allocation (11.9% expected volatility)

Expected return 7.8% compound vs. 7.4% required

55% probability of preserving real value over 20 years

Decisions requested: allocation, ranges, benchmark, rebalancing policy

The Endowment's mandate

Fixed client profile

- $2.0bn, perpetual horizon, liquid ETFs only (11-fund universe)
- Spending: 4.5% a year of the trailing 12-quarter average market value
- Costs: 0.40% a year → return objective inflation + 4.9% = 7.4% nominal (compound)
- Risk limit: expected volatility ≤ 12% and ≤ 5% probability of a one-year loss worse than −20%
- Constraints: max 40% per ETF; equities 40–70%; Treasuries + TIPS ≥ 15%; gold + commodities ≤ 10%
- Goal: intergenerational equity — future students receive the same real support as today's

Capital market assumptions

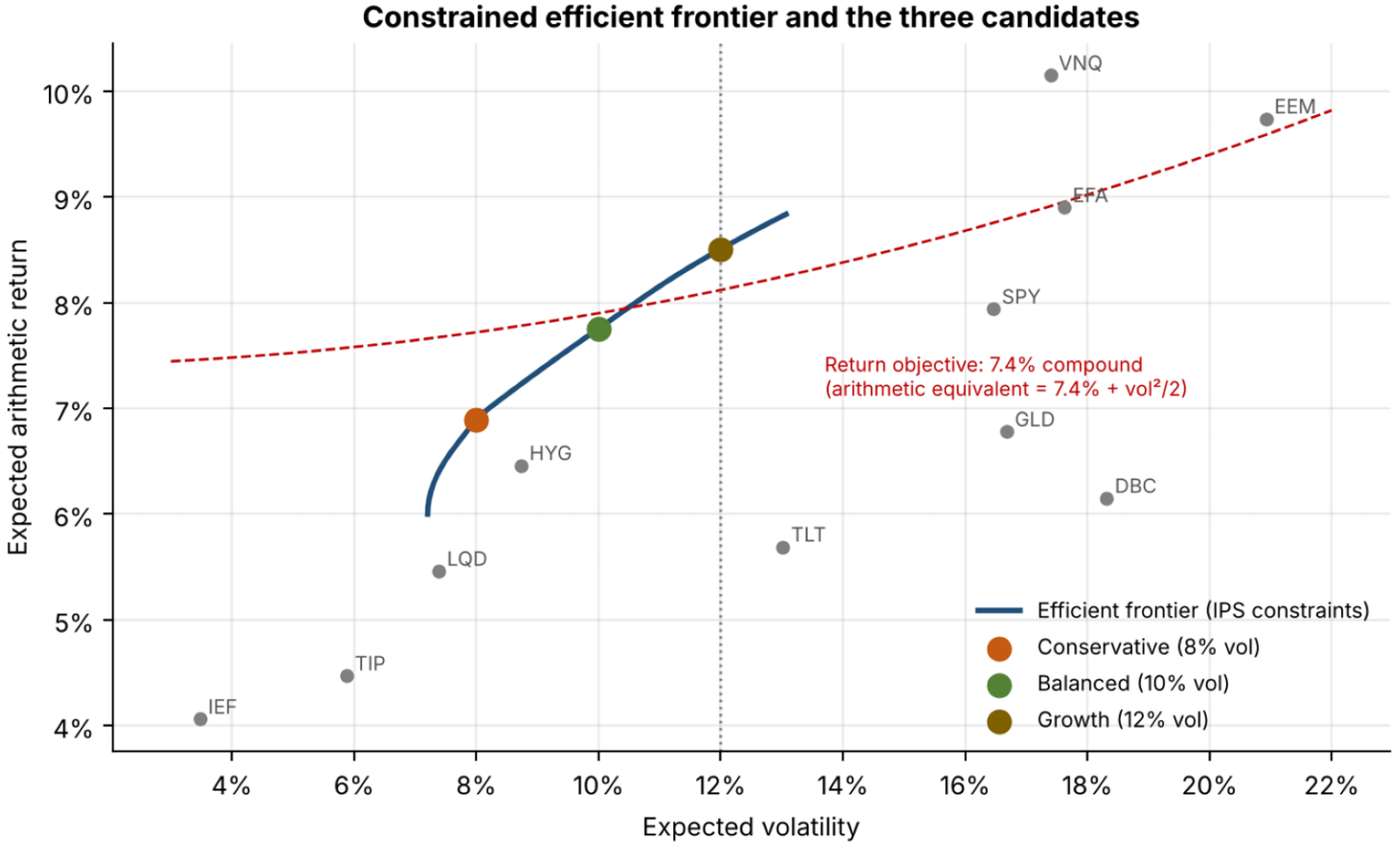
J.P. Morgan Asset Management 2026 LTCMA, U.S. dollar, 10–15 years

	ETF	Expected return (compound)	Volatility
U.S. Large Cap	SPY	6.7%	16.5%
EAFE Equity	EFA	7.5%	17.6%
Emerging Markets Equity	EEM	7.8%	20.9%
U.S. Intermediate Treasuries	IEF	4.0%	3.5%
U.S. Long Treasuries	TLT	4.9%	13.0%
TIPS	TIP	4.3%	5.9%
U.S. Inv Grade Corporate Bonds	LQD	5.2%	7.4%
U.S. High Yield Bonds	HYG	6.1%	8.7%
U.S. REITs	VNQ	8.8%	17.4%
Gold	GLD	5.5%	16.7%
Commodities	DBC	4.6%	18.3%

Inflation 2.5% · Cash 3.1% · REITs and emerging markets have the highest expected returns

The efficient frontier under the IPS constraints

Three candidates at 8%, 10% and 12% expected volatility

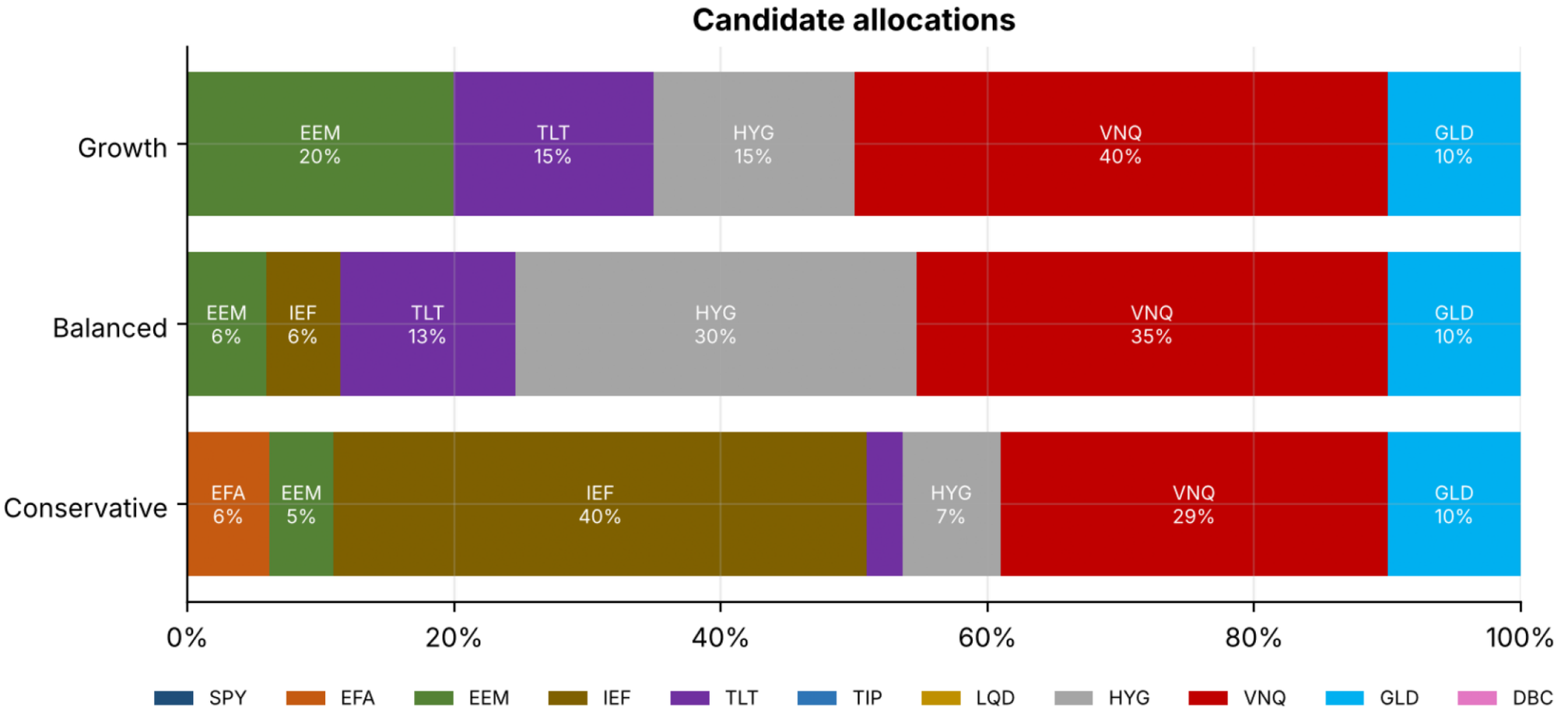


Source: J.P. Morgan Asset Management, 2026 LTCMA (USD); author's calculations.

Only Growth clears the 7.4% compound hurdle (compound: Balanced 7.3%, Growth 7.8%)

Three candidate allocations

Weights from the constrained optimizer



All three hold 0% US large caps and 29–40% REITs: the optimizer follows the highest assumed returns

Monte Carlo: 10,000 paths × 20 years

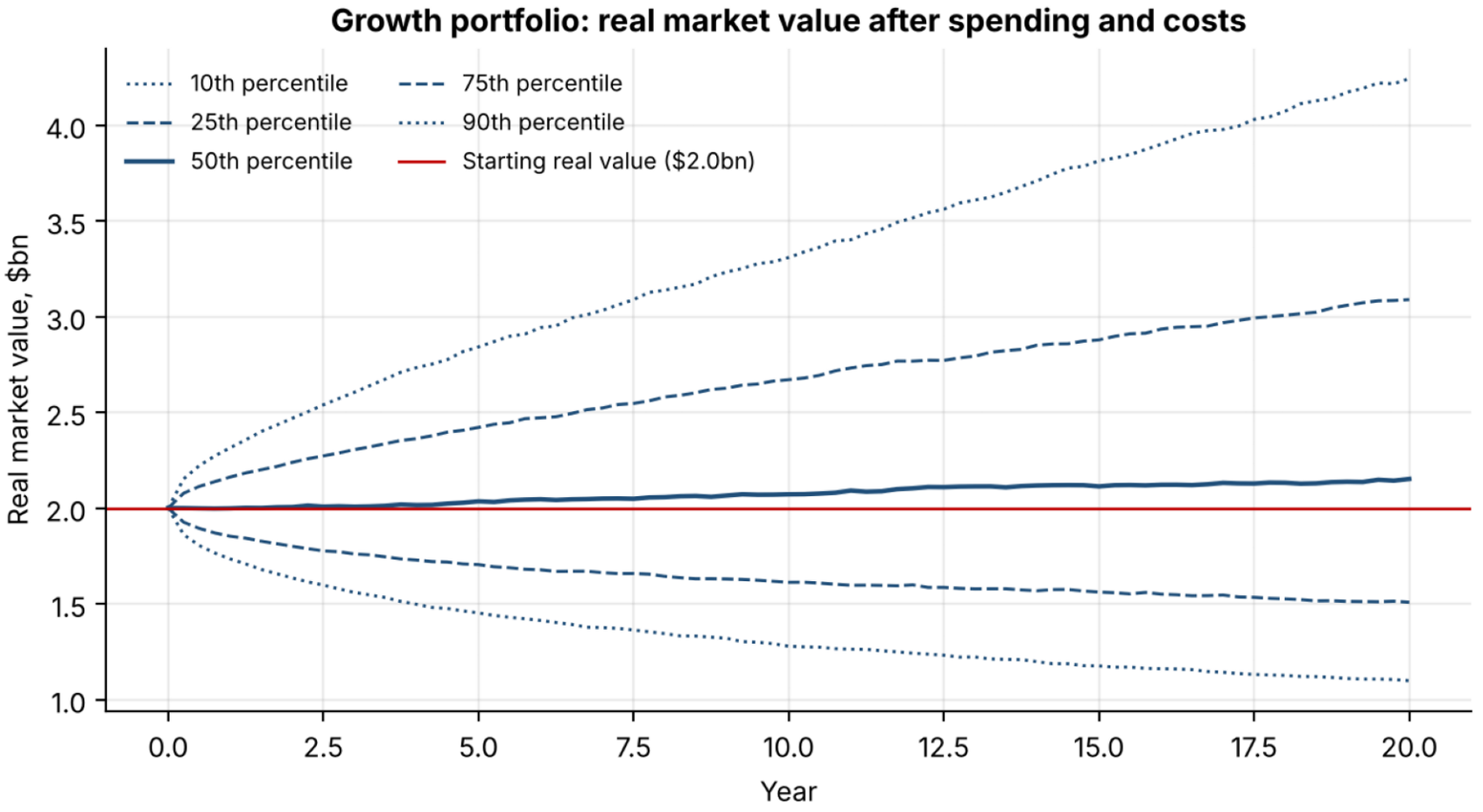
Spending rule, costs and J.P. Morgan inflation applied quarterly

	Conservative	Balanced	Growth
Expected return (arith.)	6.9%	7.8%	8.5%
Expected volatility	8.0%	10.0%	12.0%
P(real value preserved, 20y)	31.6%	47.2%	55.5%
Median real value after 20y	$1.68bn	$1.94bn	$2.15bn
P(drawdown worse than −30%)	21.9%	34.4%	48.2%
P(1-year loss < −20%)	0.00%	0.08%	0.34%
Median real spending, year 20	$74m	$84m	$93m

Growth has the highest chance of preserving real value and meets the risk limit → selected by the rule

What the recommended allocation means for spending

Growth: real market value after spending and costs, percentiles of 10,000 paths



Source: J.P. Morgan Asset Management, 2026 LTCMA (USD); author's calculations. 10,000 paths.

Median real value after 20 years $2.15bn · 48% chance of a drawdown worse than −30% along the way

Recommendation: the Growth allocation

Rounded to whole percentages within every limit; ±20% relative ranges

	Target	Range
EEM	19%	15.2% – 22.8%
TLT	15%	12.0% – 18.0%
HYG	16%	12.8% – 19.2%
VNQ	40%	32.0% – 40.0%
GLD	10%	8.0% – 12.0%

Expected return 8.5% arithmetic, 7.8% compound · volatility 11.9% · P(real value preserved) 55.2%

Risks the Committee should accept knowingly

Model limitations and what history shows

- Concentration: 40% REITs and 0% US large caps — a large bet against the market-cap-weighted world portfolio
  - It rests on one assumption (REITs 8.8% vs US large caps 6.7%): with REITs 2 points lower, the REIT weight falls from 40% to 9%
- History is harsher than the model: on 2007–2026 ETF returns this allocation had 14.0% volatility
  - Worst 12 months -37.7% (to Feb 2009); -21.9% in 2022 — the model calls a −20% year a 0.3% event
- Thin margin: expected compound return beats the 7.4% hurdle by about 37 bp; 45% chance real value is not preserved
- Mitigants: liquid ETFs, smoothed spending rule, quarterly risk report with stress tests

Implementation and decisions requested

For approval

- Policy benchmark: the target allocation in the same ETFs, rebalanced monthly
- Rebalancing: month-end check; full rebalance when a holding leaves its ±20% range (23 times in 2007–2026)
  - The IPS limits apply to targets, not drifted weights — otherwise 217 rebalances in 233 months
- Reporting: monthly to staff, quarterly risk and performance to the Committee, annual to the Board
- Annual review after new capital market assumptions; no changes on recent performance alone
- Decisions: (1) adopt the allocation and ranges, (2) adopt the benchmark, (3) adopt the rebalancing policy